

Energy Project

Thermodynamics II: MECH 351

Presented to Dr. Lyes Kadem
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1. INTRODUCTION

1.1 Current State of Energy in Africa

With the 2nd highest and most rapidly expanding population of any continent, meeting the energy needs of the United-States of Africa (USA) is a complex challenge that is set to become even tougher in the near future [1], [2]. The largest part of the continent's working population is engaged in agriculture through subsistence farming and the production of cash crops. As such, the highest population density is typically found near river basins, coastal regions, and certain highland regions, while the desert and savanna areas are generally unpopulated, as shown in Figure 1[1].

With few exceptions, the economic state of countries found in the USA is generally considered underdeveloped and significant portion of the population lacks reliable access to electricity; especially in central Africa as seen in Figure 2 [1], [3].

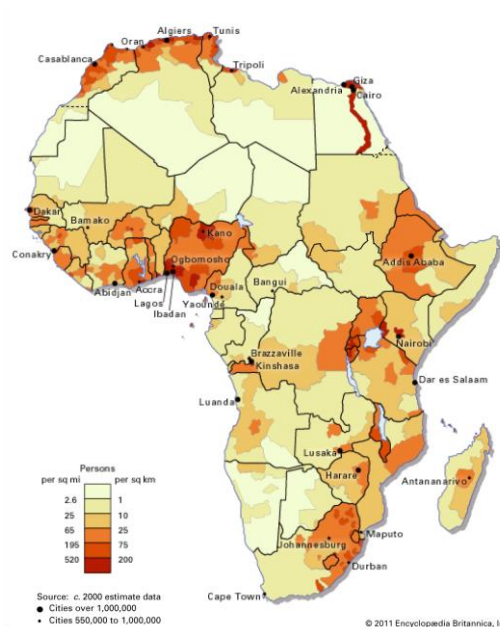


Figure 1: Population distribution across the USA [1]

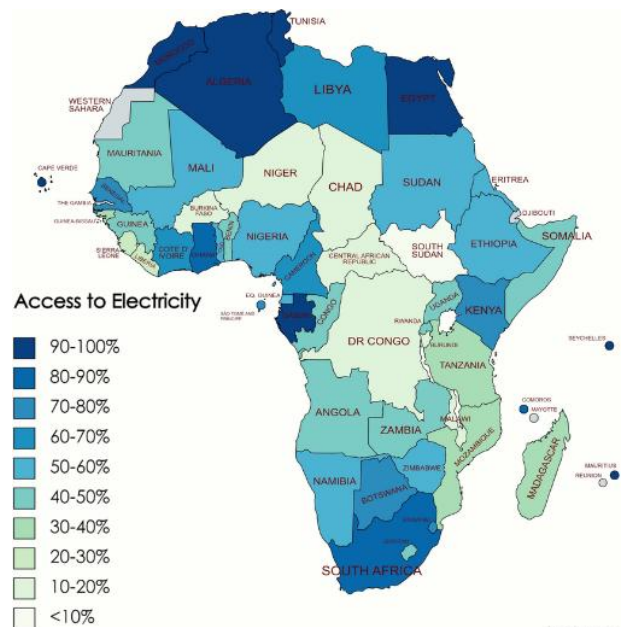


Figure 2: Access to electricity by country as a percentage of the total population [4]

This means that as the USA begins to modernize and industrialize, the need for renewable sources of energy will increase exponentially, and access to clean energy will be an essential tool in lifting various populations out of chronic states of poverty [3]. In the USA, the current energy consumption per capita is 632 kWh/capita which is extremely low compared to the consumption of 14.591 MWh/capita in Canada [5], [6]. As illustrated in the infographic Figure 3 below, hydropower generation makes up the largest source of energy production for around half of the countries in the USA with fossil fuels accounting for the other half. Eswatini and Kenya are the only exception with the largest part of their power generation comes from Bioenergy and Geothermal respectively [7].

As of 2022, almost 50% of the total final energy consumption across Africa came from residential use, with industrial and transport uses accounting for the next largest parts at 16.8% and 23.7% of total consumption, respectively [5]. This large share of power consumption in-residence is an indicator of low overall energy efficiency, and is reflected by the ratio of total energy supply per unit of economic output, or energy intensity [5]. This lower-than-average energy intensity ratio suggests that as the African economy becomes more efficient and productive in the future, their energy needs will significantly increase with their room for growth in productive energy usage.

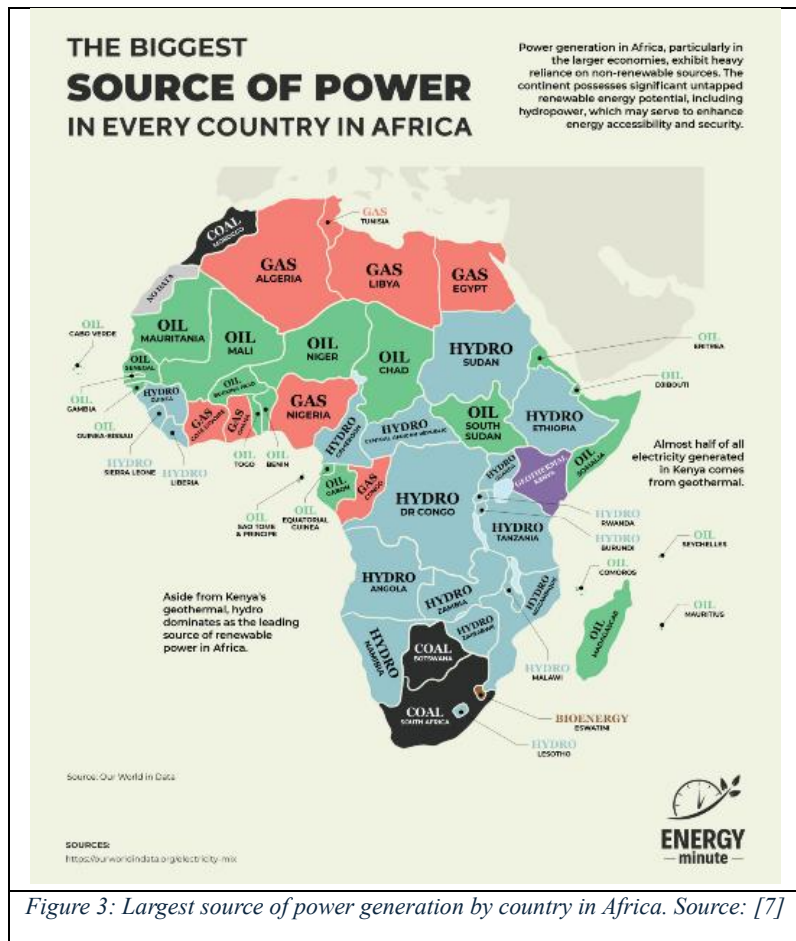


Figure 3: Largest source of power generation by country in Africa. Source: [7]

This imminent and exponential growth in energy needs across the USA means determining the most effective, clean, and renewable source of energy production is essential for fueling the growing African economy without sacrificing the long-term health and stability of the region.

1.2 Renewable Energy Map

The United-States of Africa (USA) shows an impressive diversity of weather systems, climates and patterns. Occupying over a fifth of the world's land surface and being situated in both the northern and southern hemisphere, the USA has landscapes varying from the world's largest deserts to lush rain forests. The variety of climates in the USA makes it an ideal candidate for the development of renewable energies, such as solar, wind, hydroelectricity and geothermal energy farms.

When comparing the different types of potential energy sources, each have their own sets of pros and cons. Geothermal energy is known to have many advantages over other renewables. First, geothermal energy is theoretically independent of temperature, climate or weather: its efficiency and energy output depend on the powerplant location. Moreover, geothermal energy is a very stable form of energy with a very high thermal efficiency. In fact, this type of energy used to dominate

the renewable energy market until the development of newer technologies and the rise of solar and wind farms following—when geothermal energy has mostly been cast aside. Despite this, geothermal energy shows immense potential [1]. In the case of the USA, the presence of multiple volcanoes across the continent generates a very strong opportunity for geothermal energy production. Figure 4 shows the distribution of volcanoes, hot springs and current geothermal powerplants in the USA. When comparing this to Figure 5, the immense untapped market for geothermal power in the USA is apparent. The potential high energy areas are also near large cities and large bodies of water, which would facilitate the geothermal power plant building and maintenance.

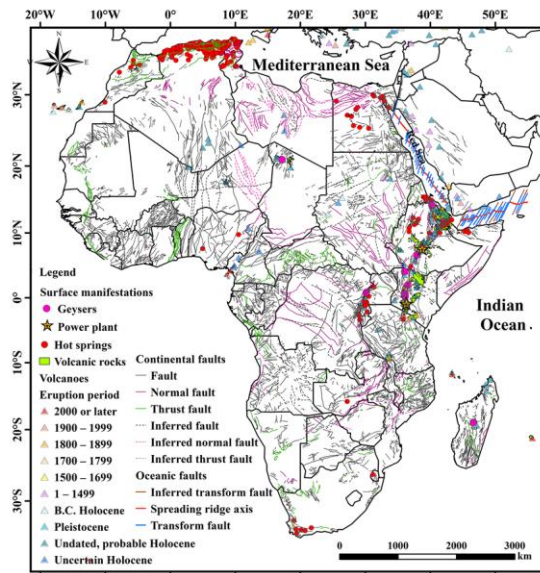


Figure 4. Distribution of volcanos obtained from USGS worldwide GIS global coverage database and installed geothermal powerplants. Source: [2].

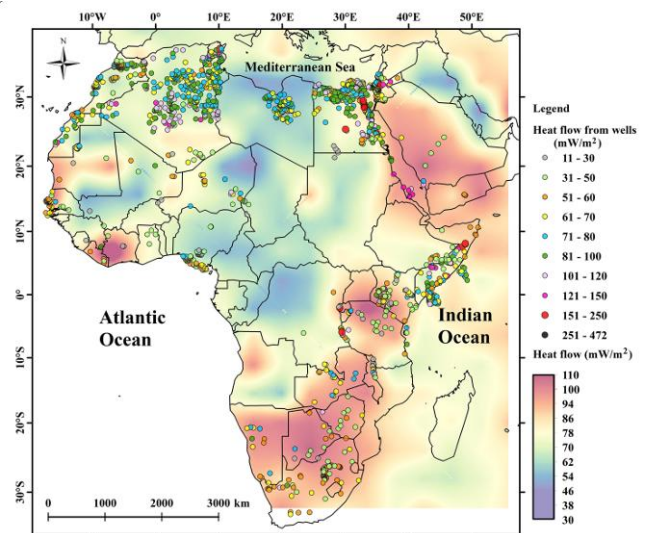


Figure 5. Heat flow map of the USA. Source: [2]

2. SCHEMATIC REPRESENTATION

To better understand how geothermal energy can be harnessed as a reliable and renewable energy source in the United States of Africa, it is essential to examine the internal workings of a typical geothermal power plant. Figure 6 illustrates the main components of the system and the flow of energy from beneath the Earth's surface to electrical output. In essence, a binary cycle is a combination of two cycles—from which it derives its name. The first cycle uses water and steam from the ground and the second uses an organic—or carbon-based—fluid.

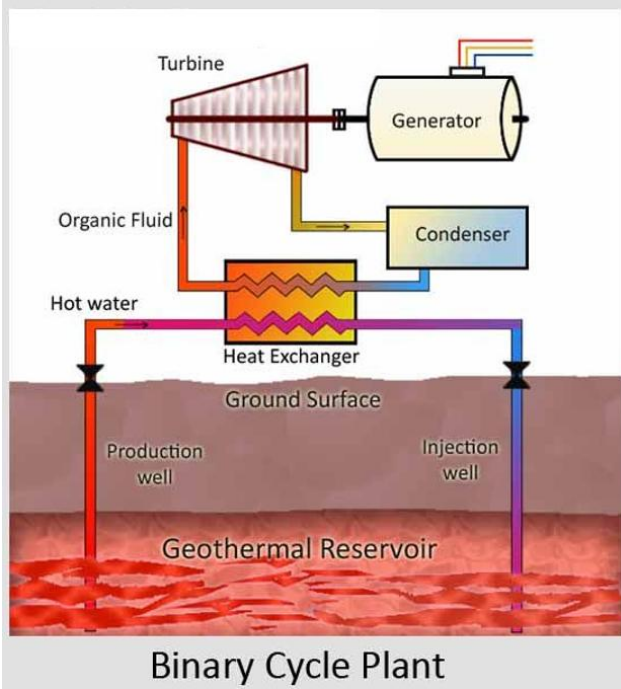


Figure 6. Visual Schematic of a Geothermal Binary Cycle Plant.
Source: [10]

First, naturally heated water and steam from the geothermal reservoir is brought to the surface, transfers its heat to a secondary, organic fluid through a heat exchanger then re-injected into the ground to repeat the cycle. The organic fluid has a lower boiling point so that it is vaporized when it absorbs heat at the heat exchanger. This vapor is then directed towards a turbine, causing it to spin. The turbine is connected to a generator, which converts the mechanical energy into electrical energy. Next, the vapor is cooled in a condenser and converted back into a liquid. This liquid then re-enters the heat exchanger, allowing the geothermal system to operate sustainably in a closed loop [8], [9].

3. ENGINEERING ANALYSIS

Unlike other geothermal systems, binary cycle plants operate using low-temperature geothermal fluids, typically below 150°C [11]. According to a meta-study investigating the effect of plant parameters on the power output and efficiency of geothermal binary cycle power plants, a higher mass flow rates produces more power. However, there is an upper limit constraining the mass flow rate, due to the well head pressure [8]. Additionally, the difference in temperature between the entrance and the exit of the heat exchanger is lower for higher mass flow rates. Equation 1 below represents the amount of heat that flows into the plant per unit of time.

Furthermore, the efficiency of these powerplants can be calculated using Equation 2. The efficiency is dependent on the power output and the heat flow into the power plant, which both increase with mass flow rate. Therefore, the efficiency will increase the higher the mass flow rate.

$$\dot{Q} = \dot{m} \times c_p \times \Delta T \quad \text{Equation 1}$$

$$\eta = \frac{\dot{W}}{\dot{Q}} = \frac{\dot{W}}{\dot{m}(h_{in} - h_{out})} \quad \text{Equation 2}$$

Another important efficiency to calculate is the efficiency of the heat exchanger. As shown in Figure 7, this efficiency is linearly dependent on the temperature difference before and after the heat exchanger. One of the reasons binary plants are so ideal is due to the fact that they do not need

high temperatures to produce geothermal power, even though it has been shown that the efficiency will increase with more heat coming from the incoming fluid [11].

In summary, maximizing the mass flow rate, as well as the heat exchanger efficiency maximizes the efficiency of a binary power plant. The greater the difference between the temperature in the inlet of the heat exchanger and the outlet, the greater the heat exchanger efficiency, leading to an overall higher efficiency.

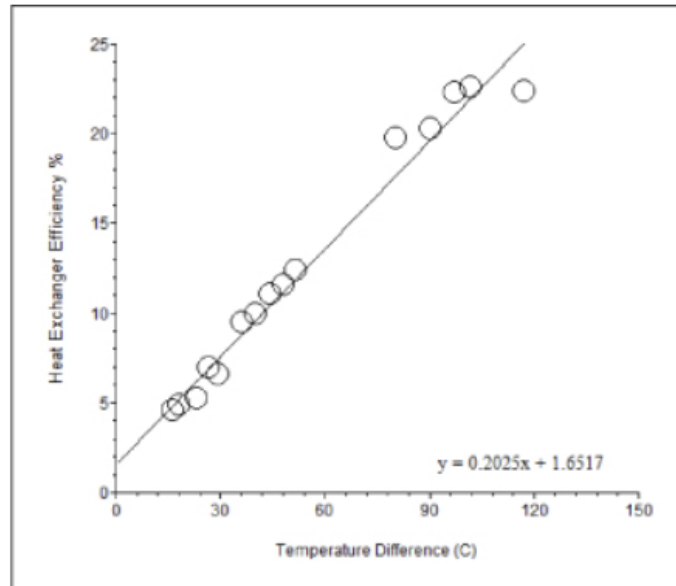


Figure 7: Heat Exchanger Efficiency Depending on the Temperature Difference

4. INFRASTRUCTURE ANALYSIS

4.1 Establishing potential

Geothermal energy requires three elements: heat source, fluid and permeability [12]. Permeability refers to the ability of fluid to move through rock. Higher permeability increases the mass flow rate and improves heat absorbance which can be converted to electricity [11]. Large-scale infrastructure such as continent-wide geothermal energy requires community projects over geothermal sources. In principle, geothermal power plants are large projects that form part of the core and then accompanying infrastructure such as high voltage transmission lines are secondary. Geothermal has two components, heating and electricity generation. Heating requires secondary infrastructure, notably underground pipelines to heat community structures. This report exclusively focuses on electricity production to sustain and meet current and future needs in the United States of Africa.

4.2 Geothermal Analysis

The traditional method for the feasibility of geothermal energy involves expensive drilling and manual field surveys. Modern technologies allow the use of thematic layers of evidence for targeted approaches, reducing capital cost and time. Elbarbary et al. developed a five-step proven thematic approach focusing on the distance to the surface, geological and structural evidence, volcanic and geothermal surface manifestation, seismic activity, and heat flow from magnetic data. This method of weighted Geographic Information System (GIS) geothermal energy potential has been used in Mexico, Indonesia and Turkey to reduce cost and enhance accessibility to this renewable energy. The largest consideration comes from

distance to surface weighted at 38.8% and the lowest is the heat flow at 9.7% [14]. The surface manifestation includes volcanoes, hot springs and geysers. These parameters most influence the GIS analysis. A case study of Turkey reveals that the predictor map found potential areas in five regions, where two have been explored and successful in installing geothermal power plants [15].

4.3 Results

The overlay of GIS thermal evidence indicates regions in the United States of Africa for potential geothermal energy exploitation. The size of 30.38 million square kilometres, poses a challenge if the potential was concentrated in one region. Fortunately, as per Figure 8 the equitable distribution between the East, North and South of potential locations allows to divide the continent into regional production corridors.

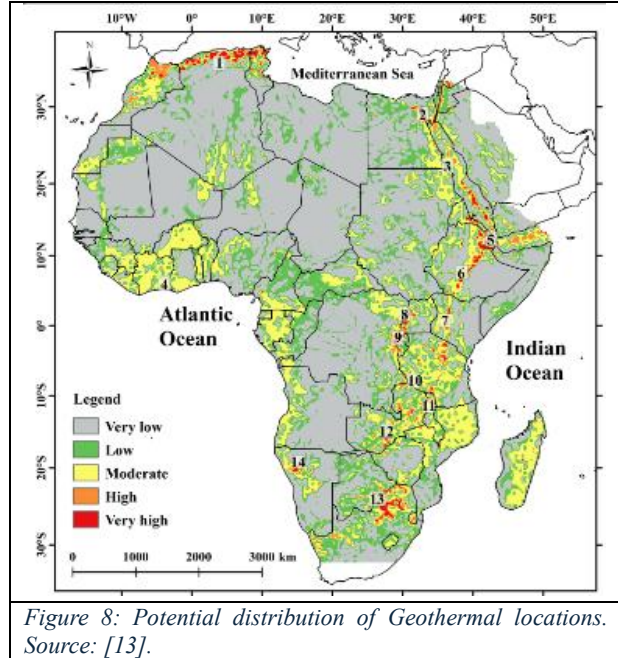


Figure 8: Potential distribution of Geothermal locations.
Source: [13].

Table 1: Regional division and properties of Africa.

Potential Region	African Region	Potential Geothermal Opportunity	Population Density Proximity
Northern Africa (1)	North	Very high	Yes
Gulf of Suez and Aqaba (Egypt) (2)	North	Very high	Yes
Red Sea Access (3)	North and East	Very high	No
Djibouti (5)	East	Very high	Yes
Main Ethiopian Rift (6)	East and Central	Very high	Yes
Kenyan Rift (7)	East and Central	Very high	Yes
Lake Albert Rift (8)	East and Central	Very high	No
Lake Kivu Rift (9)	East and Central	Very high	Yes
Tanzania (10)	Central and Southern	Very high	Yes
Northern Malawi (11)	Central and Southern	Very high	No
Central Zambia (12)	Central and Southern	Very high	No
Botswana and South Africa (13)	Southern	Very high	Yes
Namibia (14)	Southern and Central	Very high	No

The GIS compiled provides an insight into which African region and the accompanying geothermal source. By regionalizing geothermal sources, it allows for better infrastructure management by consolidating infrastructure projects and focus designs by geolocation allowing for economies of scale based on common topography and environmental factors. Moreover, as per Table 1, potential sources are accompanied by a visual confirmation of their proximity to high dense regions. An important barometer for economical feasibility of the project. West Africa does not have a high or very high potential for geothermal, and alternative resources such as wind and solar can be explored for this region.

4.4 Powerplant Selection

There are three geothermal power plant types: Binary Cycle, Single or Double Flash and Dry Steam. The specific type of power plant will be determined after drilling and field surveys. These depend on geological and thermodynamic factors at the source.

Table 2: Properties of different geothermal methods.

Parameter	Binary Cycle	Double Flash	Dry Stream
Temperature	100-200C ¹	180 to 300C	150+
Fluid Type	R-134a, isobutane ¹	Geofluid & isobutane	Geofluid
Emission	Low (closed loop) ¹	Moderate ²	Moderate ³
Efficiency	10-13% ⁴	12 to 20% ⁵	20 to 30 % ⁵
Production Range	2 to 50 MW with avg of 6 MW ⁶	2 to 110 MW ⁷	8 to 140 MW ⁸

4.5 Power Plant Construction Requirements

The construction of a geothermal plant depends on the type chosen based on geological properties. The main components found are steam supply, heat exchanger, turbine, plant pump, non-condensable gas removal and cooling tower. The major components needed for each powerplant can be found in Table 3.

Table 3: Component requirement based on powerplant type. Source: [21].

Equipment Needed	Dry Stream	Double Flash	Binary
Downhole Pumps	No	No	Yes
Steam Piping	Yes	Yes	No
Moisture separator	Yes	Yes	No
Evaporator	No	No	Yes
Condenser	Yes	Yes	Yes
Turbine (Steam)	Yes	Yes	No
Organic Vapour Turbine	No	No	Yes
Plant Pump	Yes	Yes	Yes
Steam Jet Ejector	Yes	Yes	No
Cooling tower	Yes	Yes	Possible

Depending on the source, plants can be modular and are scalable based on geological availability. Moreover, the large region available can allow for more local power plant stations for communities outside urban centres. Combined and hybrid plants are possible, and the efficiency of the performance can be assessed using the Second Law of Thermodynamics for specific situations.

¹ Source: [16], [17], [18].

² Source: [18].

³ Source: [12], [19].

⁴ Source: [11].

⁵ Source: [17].

⁶ Source: [17], [19], [20].

⁷ Source: [19].

⁸ Source: [12], [13].

Advances in turbines, pumps, and heat exchangers can improve thermal efficiencies with future upgrades.

Beyond power, an important secondary structure to consider is transmission lines to respond to the transportation of energy outside geothermal zones towards urban and rural communities. A high voltage direct current (HVDC) primary grid will allow for the reduction of transmission losses. The installation of electric accumulators can ensure a reduction of losses during peak consumption periods, which will absorb peak loads, avoiding straining the HVDC grid. HVDC has been implemented in southern and eastern Africa, allowing it to integrate with the new technology being developed. HVDC offers grid stability by allowing automation control in power flow and integrating other renewable energy [22]. The HVDC transports DC using inverters, and the DC can be converted to AC to supply local communities.

5. FUTURE DEVELOPMENTS OF GEOTHERMAL ENERGY

To ensure that the United States of Africa achieves its goal of generating 50% more of its electricity from renewable sources by 2050, it is important to ensure that the infrastructure can withstand several decades of use, as well as allowing for future expansion and modifications. Considerations of the future of geothermal energy and its infrastructure, policies surrounding renewable energies and the environmental impact of the new developments must be considered to ensure long-term sustainability, efficiency and reliability. This section will focus on the scalability and expansion of geothermal energy, the possibility of modernization, the importance of economics and policy trends in sustainable energy developments and, finally, the long-term environmental and social impacts of the proposed geothermal energy project in the USA.

5.1 Scalability and Expansion

To ensure the success of the proposed geothermal energy project, the infrastructure must be modifiable and upgradable to allow for future expansions.

5.1.1 Regional Expansion

Initial infrastructure implementation should be decided based on resource availability and geothermal potential. It is also important to take into consideration the population density and general climate of the selected area. To implement new renewable energy projects that can become hybrid projects, the area should be decided based on resource availability. Sections 0 and 4.3 outline the GIS results showing the areas that represent the highest potential for high geothermal energy outputs. Following these results, geothermal energy powerplants will be mainly introduced in the western regions of Africa. These regions are densely populated and are close to large bodies of water. They are also in areas that have a diverse range of climates, allowing for future expansion of other renewable energy sources.

5.1.2 *Hybrid Renewable Systems*

As mentioned in the section above, to meet increasing demand in renewable energies, other projects based on resource availability in specific regions should be explored, such as solar farms in the Sahara region (central USA) and wind farms in coastal and high-altitude regions (northern USA) [3]. Combining various energy sources to create hybrid renewable energy systems allow for higher reliability and reduce issues due to infrastructure damage or unexpected temporary weather conditions. Examples of hybrid renewable systems are:

1. **Solar energy in the Sahara and Sahel regions**, which have a high concentration of solar radiation, with geothermal energy storage systems.
2. **Hydropower in the coastal regions and riverine regions**, such as central USA along the Congo river or eastern USA along the Nile basin to store excess solar energy.
3. **Wind Power in high altitude and coastal regions**, such as northern USA and the Ethiopian highlands and solar energy to help with wind and sun variations between day and night.

5.1.3 *Cross-Border Energy Cooperation and Decentralized Energy Production*

It is also vital to consider the importance of cross-border cooperation. By implementing interconnected renewable energy systems, different parts of the USA can provide energy to the parts that have energy shortages. This can lead to an increase in economic activity between areas and African states. Moreover, this will help energy be distributed more equitably between all areas as well as support off-grid systems for communities living far from densely populated cities.

5.2 **Economic and Policy Developments**

To ensure the energy market remains prosperous, we recommend implementing government incentives for people, companies and neighbouring countries to participate in future projects. The government will play an important role in ensuring the renewable energy sector works in the USA. This can be done through subsidies for renewable energy projects such as the geothermal powerplants, solar and wind farms and hydroelectric plants. This can entice companies and individuals to invest in new energy projects by allowing tax deductions. It would also be pertinent to implement a net metering policy to allow households and businesses that have their own solar panels to sell their own excess electricity.

Moreover, if the government helps in the renewable energy market, this will lead to more jobs for the people living close to the power plants. These jobs would be in the construction and development of the powerplants, the maintenance and other companies that would work closely to the energy projects.

6. ENVIRONMENTAL AND SOCIAL ASSESSMENT

The implementation of geothermal systems (GS) in the USA implicates important advantages and disadvantages that must be considered. Most notably, this engineering consulting firm has found that 50% of the USA's energy production can be generated using geothermal systems. This solution does however carry with it significant political risks and logistical challenges.

The advantages of GS are evident: This technology promises reduced greenhouse gas emissions compared to biomass, a reduction in the physical footprint required for energy production as energy capture can occur mostly underground. As mentioned, low temperature geothermal power generation, or enhanced geothermal systems (EGS) can be implemented nearly anywhere, and can excess generated by the plant in rural areas can be reused for municipal purposes [23].

Additionally, significant political and military tensions exist in the African continent regarding the limited natural distribution. For example, in the Niger Delta where oil workers—including Canadians—have been kidnapped by Nigerian armed groups demanding greater shares of oil profit [24]. An increase in renewable and more widely available energy sources in the area may help dissipate violent demand around this more contentious resource.

There are also environmental and social risks associated with GS. Notably, there may be violent or political elite backlash to the attempted desaturation of the oil industry. Distribution is also a significant challenge, as the majority of energy generation related pollution in Africa is a byproduct methane rich biomass combustion in rural areas, which are the farthest from centralized energy production plants. A weakness of geothermal production is the cost effectiveness of the plant is non-viable at small scale.

Additionally, areas too dry with low capacity for heat transfer through moisture, such as in the Sahel region, may require more expensive and experimental EGS which may only be competitive in the year 2030 [23]. Finally, viable GS sites, even in regions validated by GIS is a broad and environmentally invasive process. M. Cichon writes that this process first involves risk reduction and “extensive field exploration. Exploratory wells are then drilled to determine the extent of the resource, and if successful, an injection well is planned at the site. Water is then pumped through the injection well to create and reopen fissures in the rock” during reservoir development. With these limited exploration and drilling technologies, the case-by-case implementation geothermal technologies in a widespread effort will involve significant damage to the affected environment [25].

7. CONCLUSION

In conclusion, the incorporation of GS represents a viable and strategic solution for reducing biomass-related air pollution, mitigating deforestation, and enhancing living standards in the USA. However, successful implementation will require navigating political and environmental challenges, including difficulties in rural distribution and resistance from entrenched interests in the oil sector. Our consultancy, based on the findings of this report, recommends the commission of a detailed risk assessment, political stakeholder and end-user engagement, and investment strategy, and the rolling out of a phased, regionally tailored approach to widespread implementation.

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